

GED Science Practice Test 17

PART A — QUESTIONS

GED-SCI17-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a daily hand-strength exercise affects fine-motor control. Group A squeezes a therapy ball for 5 minutes each day for four weeks. Group B continues normal activity. At the end of the study, both groups complete the same pegboard task.

Group | Average time to complete task (seconds)

A | 41

B | 54

Which conclusion is best supported by the data?

- A) The exercise group completed the fine-motor task faster on average.
- B) The normal-activity group completed the fine-motor task faster on average.
- C) Both groups completed the task in exactly the same time.
- D) Fine-motor control cannot be measured with a timed task.

GED-SCI17-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares four moving objects. Kinetic energy can be calculated as $\frac{1}{2} \times \text{mass} \times \text{speed}^2$.

Object | Mass (kg) | Speed (m/s)

A | 2 | 5

B | 4 | 4

C | 6 | 3

D | 10 | 2

Which object has the greatest kinetic energy?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI17-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A city is built near a fault where two blocks of crust move past each other. Engineers design bridges

and pipelines with flexible joints so the structures can move slightly without breaking during ground shaking.

Which hazard are the engineers mainly preparing for?

- A) Earthquakes caused by movement along a fault
- B) Lunar eclipses caused by Earth's shadow
- C) Soil formation from decayed leaves
- D) Comet tails caused by solar heating

GED-SCI17-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The pituitary gland releases hormones that help regulate other endocrine glands. One pituitary hormone signals the adrenal glands to release cortisol. Cortisol helps the body respond to stress and also feeds back to the brain and pituitary to reduce further release of the signaling hormone.

Which idea is best illustrated by this process?

- A) Hormones can regulate other hormones through feedback loops.
- B) Hormones are used only to digest fats.
- C) Cortisol prevents all cells from using energy.
- D) The pituitary gland controls only bone movement.

GED-SCI17-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A 12 kg cart moves at 4 m/s.

Kinetic energy = $\frac{1}{2} \times \text{mass} \times \text{speed}^2$

What is the cart's kinetic energy in joules?

Type your answer as a whole number.

GED-SCI17-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the height of a starting platform affects the speed of a marble at the bottom of a track. The same marble and same track surface are used in every trial.

Which TWO factors should be kept constant?

- A) Type of marble

- B) Track surface
- C) Starting platform height
- D) Speed at the bottom
- E) Final measured value

GED-SCI17-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person inherits one A allele and one B allele for blood type. Both alleles are expressed, producing type AB blood. This is an example of [D1], because both alleles affect the [D2].

Dropdowns:

D1) codominance | sublimation | erosion

D2) phenotype | orbital period | rock layer

GED-SCI17-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A bicycle uses gears to change how force and speed are transferred from the pedals to the wheel. In a low gear, the rider can climb a hill with less force at the pedals, but the pedals must turn more times for the same wheel motion.

Which statement best describes the role of the low gear?

- A) It provides mechanical advantage by trading distance moved for reduced input force.
- B) It removes friction from every part of the bicycle.
- C) It changes the rider's muscle cells into batteries.
- D) It prevents the wheels from rotating.

GED-SCI17-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A planetary scientist studies craters on two rocky surfaces. Surface A has many overlapping craters. Surface B has fewer craters and shows evidence of more recent lava flows that covered older features.

Which conclusion is best supported?

- A) Surface A is likely older than Surface B.
- B) Surface B must be older because it has fewer craters.
- C) Craters form only in oceans.
- D) Lava flows cannot change a planet's surface.

GED-SCI17-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic measures average cortisol level before and after a stress-management program.

Group | Average cortisol before ($\mu\text{g/dL}$) | Average cortisol after ($\mu\text{g/dL}$)

Stress-management group | 21 | 14

Usual-routine group | 20 | 19

Which claim is best supported by the data?

- A) The stress-management group showed a larger decrease in cortisol.
- B) The usual-routine group showed a larger decrease in cortisol.
- C) Cortisol increased in both groups.
- D) Both groups had identical cortisol changes.

GED-SCI17-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Mass placed on the platform
- B) Amount the support beam bends
- C) Same beam material used in every trial
- D) Color of the platform

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI17-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The cerebellum is a part of the brain involved in balance and coordination. It receives information from the eyes, inner ears, muscles, and joints. Damage to the cerebellum can make movements unsteady even when the muscles themselves are not weak.

Which function is mainly associated with the cerebellum?

- A) Coordinating balance and movement
- B) Filtering wastes into urine
- C) Producing bile for fat digestion
- D) Making red blood cells in bone marrow

GED-SCI17-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student rubs a glass rod with silk and then brings the rod near a thin stream of water. The water stream bends slightly toward the rod because charges in the water molecules shift in response to the charged rod.

Which interaction best explains the bending stream?

- A) Electrostatic attraction after charge separation
- B) Nuclear fusion inside water molecules
- C) Mechanical advantage from a lever
- D) Sediment compaction

GED-SCI17-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A city increases protected natural land from 3,200 hectares to 4,000 hectares.

What is the percent increase in protected natural land?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI17-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher tests whether a daily balance drill improves ankle stability. One group completes the drill for six weeks. A comparison group continues normal activity. Both groups are tested with the same ankle-stability measurement before and after the study.

Why should the same ankle-stability measurement be used for both groups?

- A) To make the results more comparable between groups
- B) To guarantee every participant improves
- C) To prevent ankle stability from changing
- D) To remove the need for a comparison group

GED-SCI17-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how different beam shapes support weight. Beams made from the same material and same total mass are shaped as a flat strip, an L-shape, a U-shape, and a box shape. Weight is added until each beam bends 1 cm.

What is the independent variable?

- A) Beam shape
- B) Weight needed to bend the beam
- C) Same material
- D) Same total mass

GED-SCI17-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A moving skateboard has kinetic energy. When friction slows the skateboard, some kinetic energy is converted mostly into [D1]. Because of this energy transfer, the skateboard's speed [D2].

Dropdowns:

D1) thermal energy | inherited DNA | groundwater

D2) decreases | increases forever | becomes zero instantly before friction acts

GED-SCI17-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A neurologist measures average balance score before and after a coordination therapy program. A higher score means better balance.

Group | Average score before | Average score after

Therapy group | 62 | 78

No-therapy group | 63 | 66

Which statement is best supported by the data?

- A) The therapy group showed a larger improvement in balance score.
- B) The no-therapy group showed a larger improvement in balance score.
- C) Both groups decreased by the same amount.
- D) Balance scores were identical after the study.

GED-SCI17-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the force needed to lift the same load using different pulley systems.

Pulley system | Input force needed (N)

A | 180

B | 120

C | 90

D | 60

Which pulley system provides the greatest mechanical advantage?

- A) System A
- B) System B
- C) System C
- D) System D

GED-SCI17-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A region tracks average spring temperature and first flowering date for a common tree species.

Year | Average spring temperature (°C) | First flowering day of year

1 | 10.2 | 116

2 | 10.8 | 112

3 | 11.4 | 108

4 | 12.1 | 103

Which TWO statements are best supported by the data?

- A) Average spring temperature increased from Year 1 to Year 4.
- B) First flowering occurred earlier in the year from Year 1 to Year 4.
- C) Average spring temperature decreased every year.
- D) First flowering occurred latest in Year 4.
- E) Both measurements stayed exactly the same.

GED-SCI17-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person has a mutation in a gene that helps repair damaged DNA. Cells with unrepaired DNA damage may be more likely to divide abnormally. The mutation does not guarantee cancer, but it can increase risk.

Which statement is best supported?

- A) A gene involved in DNA repair can affect disease risk.
- B) DNA repair genes are found only in plants.
- C) Mutations always prevent cells from dividing.
- D) Cancer risk cannot be influenced by inherited genes.

GED-SCI17-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student observes a skateboarder moving up a ramp. As the skateboarder rises, speed decreases and height increases. Some kinetic energy changes into gravitational potential energy.

Which energy change is mainly described?

- A) Kinetic energy changing into gravitational potential energy
- B) Thermal energy changing into chromosome number
- C) Light energy changing into groundwater
- D) Sound energy disappearing without transfer

GED-SCI17-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Basaltic lava often flows more easily than silica-rich lava. A volcano with low-viscosity basaltic lava may produce broad, gently sloping flows, while a volcano with more viscous lava may build steeper-sided domes.

Which property mainly controls how easily lava flows?

- A) Viscosity
- B) Humidity
- C) Genetic variation
- D) Moon phase

GED-SCI17-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A health survey includes 560 adults. Of these adults, 84 report a diagnosed thyroid disorder.

What percent of the adults report a diagnosed thyroid disorder?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI17-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how ramp surface affects the distance a block slides after leaving the ramp. The same block, ramp height, and release point are used in every trial. The block slides onto different surfaces.

Which factor is the dependent variable?

- A) Distance the block slides
- B) Surface type

- C) Same block
- D) Ramp height

GED-SCI17-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The thyroid gland releases hormones that affect metabolism. If thyroid hormone levels are too low, a person may feel tired and cold because cells release usable energy more slowly.

Which statement best describes one role of thyroid hormones?

- A) They help regulate metabolic rate.
- B) They form craters on planetary surfaces.
- C) They reflect sound waves in the ear canal.
- D) They turn bones into oxygen.

GED-SCI17-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the motion of four carts after equal pushes.

Cart | Mass (kg) | Change in speed (m/s)

- A | 2 | 6
- B | 4 | 3
- C | 6 | 2
- D | 8 | 1.5

Impulse equals mass $\times$ change in speed.

Which cart experienced an impulse of 12 kg·m/s?

- A) Cart A only
- B) Carts A and B only
- C) Carts A, B, C, and D
- D) Cart D only

GED-SCI17-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares four stars by estimated age.

Star | Estimated age (billion years)

- A | 0.8
- B | 2.4

C | 5.1
D | 9.7

Which star is estimated to be oldest?

- A) Star A
- B) Star B
- C) Star C
- D) Star D

GED-SCI17-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor records average hemoglobin level in patients before and after iron treatment.

Time | Average hemoglobin (g/dL)

Before treatment | 10.1

After 4 weeks | 11.4

After 8 weeks | 12.6

Which statement is best supported by the data?

- A) Average hemoglobin increased after iron treatment.
- B) Average hemoglobin decreased after iron treatment.
- C) Hemoglobin stayed exactly the same.
- D) Iron treatment made hemoglobin disappear.

GED-SCI17-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student uses a prism to separate white light into colors. The colors spread out because different wavelengths of visible light bend by different amounts when passing through the prism.

Which process separates the colors?

- A) Dispersion
- B) Blood filtration
- C) Plate subduction
- D) Gene expression

GED-SCI17-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether different warm-up routines affect muscle performance. Participants complete either no warm-up, light stretching, or dynamic movement before the same strength test.

Which TWO steps would improve the investigation?

- A) Use the same strength test for all participants.
- B) Include multiple participants in each warm-up group.
- C) Give each group a different strength test.
- D) Record only the best performance in the study.
- E) Change both warm-up type and testing equipment together.

GED-SCI17-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the frequency and wavelength of four waves.

Wave	Frequency (Hz)	Wavelength (m)
A	4	9
B	6	6
C	9	4
D	12	3

Wave speed = frequency $\times$ wavelength.

Which waves have the same speed?

- A) Waves A, B, C, and D
- B) Waves A and B only
- C) Waves C and D only
- D) Wave D only

GED-SCI17-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A damaged coral reef still contains some living coral colonies and hard reef structure. Over time, young corals and algae attach to open surfaces, and small animals return to the habitat.

Which ecological process is most directly described?

- A) Secondary succession
- B) Radioactive decay
- C) Sublimation
- D) Contact metamorphism

GED-SCI17-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A mutation changes the shape of a receptor on the surface of a cell. The receptor no longer binds its normal hormone well, so the cell responds weakly to the hormone signal.

Which statement best explains the effect?

- A) A gene change can alter receptor structure and cell signaling.
- B) Hormone signals are unrelated to cell receptors.
- C) Receptors are made only of rock minerals.
- D) Mutations always improve hormone binding.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI17-001 — Correct: A — Correct Answer: The exercise group completed the fine-motor task faster on average. — Explanation: Group A had the lower average completion time, 41 seconds, compared with 54 seconds for Group B.

GED-SCI17-002 — Correct: B — Correct Answer: Object B — Explanation: Kinetic energy values are $A = 25 \text{ J}$, $B = 32 \text{ J}$, $C = 27 \text{ J}$, and $D = 20 \text{ J}$, so Object B is greatest.

GED-SCI17-003 — Correct: A — Correct Answer: Earthquakes caused by movement along a fault — Explanation: Flexible joints help structures tolerate ground shaking from fault movement.

GED-SCI17-004 — Correct: A — Correct Answer: Hormones can regulate other hormones through feedback loops. — Explanation: Cortisol feeds back to reduce further signaling, showing hormonal feedback.

GED-SCI17-005 — Correct: 96 — Tolerance: exact — Correct Answer: 96 — Explanation: Kinetic energy = $\frac{1}{2} \times 12 \times 4^2 = 96 \text{ J}$.

GED-SCI17-006 — Correct: A,B — Correct Answer: Type of marble | Track surface — Explanation: These factors should stay constant so starting height is the main tested variable.

GED-SCI17-007 — Correct: D1=codominance; D2=phenotype — Correct Answer: D1=codominance; D2=phenotype — Explanation: In codominance, both alleles are expressed in the observable trait.

GED-SCI17-008 — Correct: A — Correct Answer: It provides mechanical advantage by trading distance moved for reduced input force. — Explanation: Low gear reduces input force but requires more pedal motion.

GED-SCI17-009 — Correct: A — Correct Answer: Surface A is likely older than Surface B. — Explanation: Older surfaces usually have more accumulated craters, while newer lava flows can cover older craters.

GED-SCI17-010 — Correct: A — Correct Answer: The stress-management group showed a larger decrease in cortisol. — Explanation: The stress-management group decreased by $7 \text{ } \mu\text{g/dL}$, while the usual-routine group decreased by $1 \text{ } \mu\text{g/dL}$.

GED-SCI17-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Mass placed on the platform; T2=Amount the support beam bends; T3=Same beam material used in every trial — Explanation: Mass is changed, beam bending is measured, and beam material is controlled.

GED-SCI17-012 — Correct: A — Correct Answer: Coordinating balance and movement — Explanation: The cerebellum helps coordinate balance and movement.

GED-SCI17-013 — Correct: A — Correct Answer: Electrostatic attraction after charge separation — Explanation: The charged rod shifts charges in water molecules, causing attraction.

GED-SCI17-014 — Correct: 25 — Tolerance: exact — Correct Answer: 25 — Explanation: The increase is 800 hectares. $800 \div 3,200 \times 100 = 25\%$.

GED-SCI17-015 — Correct: A — Correct Answer: To make the results more comparable between groups — Explanation: Using the same measurement helps researchers compare group results fairly.

GED-SCI17-016 — Correct: A — Correct Answer: Beam shape — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI17-017 — Correct: D1=thermal energy; D2=decreases — Correct Answer: D1=thermal energy; D2=decreases — Explanation: Friction converts some kinetic energy into thermal energy, reducing speed.

GED-SCI17-018 — Correct: A — Correct Answer: The therapy group showed a larger improvement in balance score. — Explanation: The therapy group improved by 16 points, compared with 3 points for the no-therapy group.

GED-SCI17-019 — Correct: D — Correct Answer: System D — Explanation: For the same load, the system requiring the least input force has the greatest mechanical advantage.

GED-SCI17-020 — Correct: A,B — Correct Answer: Average spring temperature increased from Year 1 to Year 4. | First flowering occurred earlier in the year from Year 1 to Year 4. — Explanation: Temperature increased from 10.2°C to 12.1°C, while flowering day decreased from day 116 to day 103, meaning earlier flowering.

GED-SCI17-021 — Correct: A — Correct Answer: A gene involved in DNA repair can affect disease risk. — Explanation: Reduced DNA repair can increase the chance of abnormal cell division and disease risk.

GED-SCI17-022 — Correct: A — Correct Answer: Kinetic energy changing into gravitational potential energy — Explanation: As height increases and speed decreases, kinetic energy changes into gravitational potential energy.

GED-SCI17-023 — Correct: A — Correct Answer: Viscosity — Explanation: Viscosity describes resistance to flow and affects how easily lava moves.

GED-SCI17-024 — Correct: 15 — Tolerance: exact — Correct Answer: 15 — Explanation: $84 \div 560 \times 100 = 15\%$.

GED-SCI17-025 — Correct: A — Correct Answer: Distance the block slides — Explanation: The dependent variable is the measured distance after changing the surface type.

GED-SCI17-026 — Correct: A — Correct Answer: They help regulate metabolic rate. — Explanation: Thyroid hormones influence how quickly cells use energy.

GED-SCI17-027 — Correct: C — Correct Answer: Carts A, B, C, and D — Explanation: Each cart has $\text{mass} \times \text{change in speed} = 12 \text{ kg}\cdot\text{m/s}$.

GED-SCI17-028 — Correct: D — Correct Answer: Star D — Explanation: Star D has the greatest estimated age, 9.7 billion years.

GED-SCI17-029 — Correct: A — Correct Answer: Average hemoglobin increased after iron treatment. — Explanation: Average hemoglobin rose from 10.1 to 12.6 g/dL.

GED-SCI17-030 — Correct: A — Correct Answer: Dispersion — Explanation: Dispersion separates light into colors because different wavelengths bend by different amounts.

GED-SCI17-031 — Correct: A,B — Correct Answer: Use the same strength test for all participants. | Include multiple participants in each warm-up group. — Explanation: Consistent testing and enough participants improve reliability.

GED-SCI17-032 — Correct: A — Correct Answer: Waves A, B, C, and D — Explanation: Each wave has speed 36 m/s because $\text{frequency} \times \text{wavelength} = 36$.

GED-SCI17-033 — Correct: A — Correct Answer: Secondary succession — Explanation: Secondary succession occurs after a disturbance when some biological structure or living material remains.

GED-SCI17-034 — Correct: A — Correct Answer: A gene change can alter receptor structure and cell signaling. — Explanation: Genes provide instructions for proteins, including receptors, so mutations can affect signaling.